

Functional genetic report · longevity

# **Stroke and Cerebral Small Vessels**

Sérgio F · Report date: 03-10-2026

Small vessels of the brain (arterioles and capillaries) age before there are symptoms: white matter lesions, silent lacunar infarcts and dilated perivascular spaces, which are linked to cognitive decline and dementia. It complements the stroke section of the advanced cardiovascular report. Only genes (chip + local imputation + polygenic scores).

# Contents

---

|                                                 |    |
|-------------------------------------------------|----|
| • Summary                                       | 3  |
| • How to read this report                       | 4  |
| • Polygenic scores                              | 5  |
| • Small vessel disease and white matter lesions | 6  |
| • Perivascular spaces and glymphatic system     | 7  |
| • Ischaemic stroke by subtype                   | 8  |
| • Brain hemorrhage and amyloid angiopathy       | 9  |
| • Mechanisms of ageing                          | 10 |
| • Practical suggestions                         | 11 |
| • Limitations                                   | 12 |
| • Methods and sources                           | 13 |

## Summary

**11**

SNPs analyzed

**5**

in the chip

**6**

imputed (DR2 $\geq$ 0.8)

**0**

SNP not found in the data

**5**

Polygenic scores

The disease of small brain vessels causes about 25% of ischaemic stroke and most deep hemorrhages, and contributes to almost half of dementia. Its marks on MRI are white matter hyperintensities, lacunar infarctions, microbleaching and dilated perivascular spaces. These spaces are part of the glymphatic system, the drainage pathway of the cerebral fluid that is more active during deep sleep. Common genetics pass through the basal membrane of vessels (COL4A1/COL4A2), pericytes (FOXF2), 17q25 locus and blood-brain barrier.

In summary: mixed picture. White matter lesions have a high genetic score (the 92nd percentile, already seen in the brain and neurodegeneration report), but the two strongest loci of these lesions are favourable (17q25 TT, SH3PXD2A CC) and the lacunar stroke scores (the 14th percentile), global stroke (19), ischemic stroke (35) and cerebral hemorrhage (18) are low. The weakest point is the perivascular spaces (SLC13A3 CC and LPAR1 GA), linked to glymphatic drainage, which is favored by sleep. APOE  $\epsilon$ 3/ $\epsilon$ 3.

### Favourable

- **Low stroke scores:** Stroke gap and small vessels at the 14th percentile, global stroke at the 19th percentile, ischaemic stroke at the 35th percentile, intracerebral hemorrhage at the 18th percentile.
- **17q25 (TRIM65) rs3744028 TT** and **SH3PXD2A rs7909791 CC:** without the risk alleles of the two strongest loci of white matter lesions.
- **APOE  $\epsilon$ 3/ $\epsilon$ 3** (without  $\epsilon$ 4 or  $\epsilon$ 2, alleles of amyloid angiopathy and lobar hemorrhage) and TSPAN2 CC (without the large artery stroke allele).

### Unfavourable

- **White matter lesions: 355 variant score at the 92nd percentile** (brain report and neurodegeneration).
- **Perivascular spaces:** SLC13A3 rs2425884 CC (2 risk alleles, the strongest locus) and LPAR1 rs10817108 GA; WNT7A GC.
- **1q22 (PMF1) rs2984613 CC:** homozygous for the common allele of deep hemorrhage and lacunar stroke; COL4A2 rs9515201 AC.

**Interpretation:** genetic risk of low stroke, but with a genetic trend for more white matter lesions and for less efficient perivascular drainage. The strongest modifiable factors of the small vessels are sleep, physical activity, glycemia and the absence of tobacco and alcohol; almost all are already optimized in the plan.

## How to read this report

---

**Genetics only.** All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

**Strand:** All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 5 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 2 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

**Origin and trust:** chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with  $DR2 \geq 0.8$  (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

**Effect and judgment allele:** The comparison between the number of Sérgio F copies and the mean of the reference population ( $2 \times$  allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

**Polygenic scores (PGS):** add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage  $\geq 80$  in a risk score = unfavourable;  $\leq 20$  = favourable.

## Polygenic scores

The white matter lesion score (PGS001534) was reused from the brain and neurodegeneration report, that of ischaemic stroke (PGS000665, 32 variants) from a previous calculation of the profile and that of global stroke (PGS000038) from the advanced cardiovascular report. Intracerebral hemorrhage (26 024 variants) was calculated in this report. The stroke lacunar and small vessel score uses 8 loci from GWAS Catalog (excluding studies from other populations). The score of perivascular spaces had no published effect sizes and was replaced by individual SNPs.

| Trait                                                                | Score     | Variables used (coverage) | Percentage (reference population) | Judgement              |
|----------------------------------------------------------------------|-----------|---------------------------|-----------------------------------|------------------------|
| Volume of white matter hyperintensities (355 variants)               | PGS001534 | 301 of 355 (85%)          | <b>p92</b>                        | Unfavourable           |
| Stroke lacunar and small vessel disease (GWAS Catalog score, 8 loci) | C_LACUNAR | 8 of 8 (100%)             | <b>p14</b>                        | Favourable             |
| Stroke (90 variants)                                                 | PGS000038 | 84 of 90 (93%)            | <b>p19</b>                        | Favourable             |
| Ischaemic stroke (32 variants)                                       | PGS000665 | 30 of 32 (94%)            | <b>p35</b>                        | Typical / no judgement |
| Intracerebral hemorrhage (26 024 variants)                           | PGS019222 | 25 328 of 26 024 (97%)    | <b>p18</b>                        | Favourable             |

# Small vessel disease and white matter lesions

White matter hyperintensities are areas of demyelination and axonal loss by chronic hypoperfusion and leakage from the blood-brain barrier; its volume doubles the risk of stroke and increases the risk of dementia.

**Polygenic scores of this theme:** Volume of white matter hyperintensities (355 variants) (PGS001534): the 92nd percentile Unfavourable; Stroke lacunar and small vessel disease (GWAS Catalog score, 8 loci) (C\_LACUNAR): the 14th percentile Favourable

| Gene                       | SNP        | Genotype (+) | Source                 | Effect allele | Copies | Freq. | Effect                                                                                                                                                 | Judgement              |
|----------------------------|------------|--------------|------------------------|---------------|--------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| COL4A2                     | rs9515201  | AC           | imputation<br>DR2=1.00 | A             | 1      | 0.32  | IV collagen from the basal membrane of small vessels: Allele A increases the lacunar stroke (OR ~1.12) and deep cerebral hemorrhage. (strong evidence) | Typical / no judgement |
| TRIM65/<br>TRIM47<br>17q25 | rs3744028  | TT           | imputation<br>DR2=0.99 | C             | 0      | 0.20  | Locus 17q25, the strongest of white matter hyperintensities: C allele, less common, increases the burden of injuries. (strong evidence)                | Favourable             |
| SH3PXD2A                   | rs7909791  | CC           | chip                   | A             | 0      | 0.36  | SH3PXD2A (podosomes, matrix invasion): allele A increases white matter hyperintensities. (moderate evidence)                                           | Favourable             |
| FOXF2                      | rs12204590 | TT           | chip                   | —             | —      | —     | FOXF2, transcription factor of the cerebral pericytes: locus of the stroke of small vessels and lesions of the white matter. (moderate evidence)       | Typical / no judgement |

**Favourable**

- TRIM65 (17q25) rs3744028 TT
- SH3PXD2A rs7909791 CC
- Stroke score gap at the 14th percentile

**Unfavourable**

- Score of white matter lesions at the 92nd percentile
- COL4A2 rs9515201 AC

**Interpretation:** The stronger loci isolates are favourable, but the 355 variant score is high: the load is spread by many small loci. The gap stroke score is low, which suggests that the genetic risk is more of silent lesions than of infarcts. FOXF2 TT is information (the effect allele is not well defined).

# Perivascular spaces and glymphatic system

Perivascular spaces surround the penetrating arterioles and are the pathway of entry and exit of the brain fluid into the glymphatic system, which removes proteins such as beta-amyloid especially during deep sleep. When they are dilated on MRI, they indicate less efficient drainage and are associated with small vessel disease and dementia.

| Gene    | SNP        | Genotype (+) | Source                 | Effect allele | Copies | Freq. | Effect                                                                                                                                                                 | Judgement              |
|---------|------------|--------------|------------------------|---------------|--------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| SLC13A3 | rs2425884  | CC           | chip                   | C             | 2      | 0.55  | SLC13A3 (dicarboxylate transporter in the blood-brain barrier): the strongest locus of dilated perivascular spaces; the C allele increases the load. (strong evidence) | Unfavourable           |
| WNT7A   | rs13079464 | GC           | imputation<br>DR2=0.97 | C             | 1      | 0.43  | WNT7A, signal of formation and maintenance of the blood-brain barrier: C allele increases perivascular spaces. (moderate evidence)                                     | Typical / no judgement |
| LPAR1   | rs10817108 | GA           | imputation<br>DR2=1.00 | A             | 1      | 0.20  | Receptor of lysophosphatidic acid 1: the allele A, less common, increases perivascular spaces. (moderate evidence)                                                     | Unfavourable           |

Favourable

-

Unfavourable

- SLC13A3 rs2425884 CC (2 risk alleles)
- LPAR1 rs10817108 GA (less common allele)
- WNT7A rs13079464 GC

**Interpretation:** Genetic trend for more dilated perivascular spaces. Deep sleep (slow waves) is the main moment of glymphatic drainage; melatonin is already in the plan and regular sleep is the main lever.

# Ischaemic stroke by subtype

Ischaemic stroke is divided into large artery (carotid atherosclerosis and cerebral arteries), cardioembolic artery (mainly atrial fibrillation) and small vessels (lacunar).

**Polygenic scores of this theme:** Stroke (90 variants) (PGS000038): the 19th percentile Favourable ;  
Ischaemic stroke (32 variants) (PGS000665): the 35th percentile Typical / no judgement

| Gene   | SNP        | Genotype (+) | Source | Effect allele | Copies | Freq. | Effect                                                                                                                  | Judgement              |
|--------|------------|--------------|--------|---------------|--------|-------|-------------------------------------------------------------------------------------------------------------------------|------------------------|
| TSPAN2 | rs12122341 | CC           | chip   | G             | 0      | 0.26  | TSPAN2: G allele increases large artery stroke (carotid atherosclerosis and intracranial arteries). (moderate evidence) | Typical / no judgement |

### Favourable

- TSPAN2 rs12122341 CC (without the large artery G allele)
- Stroke global at the 19th percentile and ischaemic at the 35th percentile
- Atrial fibrillation at the 2nd percentile (rhythm report)

### Unfavourable

- HDAC9 and 9p21 seen in other reports

**Interpretation:** Low genetic risk in the three subtypes: cardioembolic is low by atrial fibrillation (the 2nd percentile), large artery by TSPAN2 and low LDL, and the gap by the score at the 14th percentile.

## Brain hemorrhage and amyloid angiopathy

Deep cerebral hemorrhage is mainly caused by small vessels arteriolosclerosis; lobar, by amyloid angiopathy (beta-amyloid deposition on the wall), very linked to APOE ε4 and ε2.

**Polygenic scores of this theme:** Intracerebral hemorrhage (26 024 variants) (PGS019222): the 18th percentile

Favourable

| Gene             | SNP       | Genotype (+) | Source                 | Effect allele | Copies | Freq. | Effect                                                                                                                                     | Judgement              |
|------------------|-----------|--------------|------------------------|---------------|--------|-------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| PMF1/SEMA4A 1q22 | rs2984613 | CC           | imputation<br>DR2=1.00 | C             | 2      | 0.66  | Locus 1q22: the C allele (common) increases deep cerebral hemorrhage (OR ~1.3), lacunar stroke and white matter lesions. (strong evidence) | Unfavourable           |
| APOE ε4          | rs429358  | TT           | imputation<br>DR2=0.99 | C             | 0      | 0.15  | APOE ε4: amyloid angiopathy and lobar hemorrhage, in addition to Alzheimer's. (strong evidence)                                            | Favourable             |
| APOE ε2          | rs7412    | CC           | chip                   | T             | 0      | 0.07  | APOE ε2: protects from Alzheimer's disease but is associated with lobar hemorrhage by amyloid angiopathy. (strong evidence)                | Typical / no judgement |

### Favourable

- APOE ε3/ε3
- Intracerebral hemorrhage score at the 18th percentile

### Unfavourable

- PMF1 (1q22) rs2984613 CC

**Interpretation:** Genetic risk of low bleeding. Omega-3 in high dose and other drugs that move in coagulation have a small effect; relevant only if in the future there is an indication for anticoagulant or anti-aggregant.

# Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

| Mechanism                            | Unfavourable                                    | Favourable                   | Interpretation                         |
|--------------------------------------|-------------------------------------------------|------------------------------|----------------------------------------|
| Hematoencephalic Barrier dysfunction | White matter p92 lesions; COL4A2 AC; SLC13A3 CC | 17q25 TT; SH3PXD2A CC        | Mixed.                                 |
| Glymphatic dysfunction               | SLC13A3 CC; LPAR1 GA; WNT7A GC                  | —                            | Unfavourable; sleep is the main lever. |
| Vascular Aging                       | PMF1 CC                                         | Low stroke Scores; TSPAN2 CC | Favourable.                            |
| Protein Aggregation                  | —                                               | APOE ε3/ε3                   | Neutral.                               |
| Neuroinflammation                    | LPAR1 GA                                        | —                            | Small effect.                          |

## Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

| Topic                             | Suggestion (not applied)                                                                                                                                                                                                                   | Genetic basis            |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Deep sleep                        | Regular time, 7–9 h, dark and fresh room, and without screens in the last hour; slow wave sleep is when the glymphatic system drains more. Melatonin 0.3–1 mg at night as in the plan. <b>suggestion (not applied)</b>                     | SLC13A3 CC; LPAR1 GA     |
| Aerobic exercise                  | 150–300 min/week of aerobic exercise and strength training: it is associated with less progression of white matter lesions and better cerebral flow. <b>suggestion (not applied)</b>                                                       | White matter p92 lesions |
| Glycemia and LDL                  | Keeping glucose and ApoB low (already in the plan with dapagliflozin and retatrutide); hyperglycemia accelerates the disease of small vessels. <b>suggestion (not applied)</b>                                                             | White matter p92 lesions |
| Baseline MRI                      | Consider with the doctor a basic brain MRI (with FLAIR and T2*) to know if there are lesions of the white matter or microbleaching; serves as a reference and changes future decisions on anti-aggregants. <b>suggestion (not applied)</b> | Score of p92 lesions     |
| Hearing and cognitive stimulation | Treating any hearing loss and maintaining cognitive and social activity; both reduce the impact of small vessel lesions on cognition. <b>suggestion (not applied)</b>                                                                      | White matter lesions     |
| Coordination with the plan        | No tobacco and no alcohol. Omega-3 3.2–4 g/day (already taking fish oil). D3 10 000 IU. Curcumin, luteolin, apigenin and quercetin in pause while ALT is high. <b>suggestion (not applied)</b>                                             | V2 plan rules            |

## Limitations

---

- The monogenic forms of small vessel disease (NOTCH3/CADASIL, HTRA1, COL4A1/COL4A2 rare) are not reliably read by the chip.
- The white matter lesion score predicts volume of lesions on average; only MRI shows the real state.
- The percentiles compare with the reference population (n=633).
- Only genes: does not use image or analysis.

## Methods and sources

---

- Genotypes: Raw DNA of the chip (GRCh38, strand +), analyzed only locally; Missing SNPs (statistical imputation, public reference panel of complete 30x genomes), accepted with  $DR2 \geq 0.8$ .
- Public positions and alleles: Ensembl REST (only rsIDs). Locally calculated frequencies in the reference population ( $n=633$ ); effect alleles confirmed by the expected frequency.
- Scores: PGS019222 (intracerebral hemorrhage) calculated in this report; PGS001534 reused from the brain and neurodegeneration report and PGS000665 from a previous calculation; PGS000038 of the advanced cardiovascular report; simple score of the GWAS Catalog (local copy) for the lacunar stroke and small vessels.
- Main references: Wardlaw 2019 (small brain vessel disease); Persyn 2020 (GWAS of white matter lesions); Traylor 2021 (stroke lacunar); Mishra 2022 (GWAS of stroke, GIGASTROKE); Duperron 2023 (perivascular spaces, SLC13A3); Xie 2013 (sleep and glymphatic system); Woo 2014 (1q22 and intracerebral hemorrhage); Biffi 2010 (APOE and lobar hemorrhage).
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).